

PD4B-250-P5-LC

OVERVIEW

InGaAs PIN photodiode with extended spectral range coupled to an optical fiber and packaged into a hermetic case.

MAIN FEATURES

- Maximum optical power: 5 mW
- Typical responsivity: 1.3 A/W at 1950 nm
- Cut-off wavelength: 2150 nm
- Package types: coaxial with or without bracket, DIL-14-pins, BTF-14-pins
- Coupling technologies: R50/R30 (low back reflection)

ORDERING INFORMATION

PD4B-250-P5-LC-X-X-X-X-X-X

Coupling technology

R50: back reflection -50 dB (SM06, SM1, SM3 and SM1950 fiber, FA, SA or N connector)

R30: back reflection -30 dB (MM5, MM6, MM105 fiber)

Case type

COAX: compact coaxial (low duty cycle pulse mode only)

COAXB: compact coaxial with a bracket

DIL: standard DIL-14-pins

DILRAD: DIL-14-pins with integrated sidewall radiator

BTF: BTF-14-pins

Pinout code

7: see more details on page 5

Fiber type

SM06: SM, [Corning Hi-1060](#), furcation tubing Ø0.9 mm

SM1: SM, G.657.A1, [Corning SMF-28 Ultra](#), furcation tubing Ø0.9 mm or **BSM1** Ø0.25mm

SM3: SM, G.657.B3, [Corning ClearCurve ZBL](#), furcation tubing Ø0.9 mm or **BSM3** Ø0.25mm

SM1950: SM, [Coherent SM1950](#)

MM5: MM, [50/125.OM3](#), furcation tubing Ø0.9 mm

MM6: MM, [62.5/125.OM1](#), furcation tubing Ø0.9 mm

MM105: MM, 105/125, NA=0.22, [Coherent S105/125-22A](#), furcation tubing Ø0.9 mm

Connector type

FU: FC/UPC (MM5, MM6, MM105)

FA: FC/APC (MM5, MM6, MM105)

N: no connector (scissors cut)

Other type: on request

Fiber length

0.5: 500+/-50 mm

1.0: 1000+/-100 mm

Other length on request

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ABSOLUTE MAXIMUM RATINGS

Parameter		Value	Unit	Conditions
Maximum optical input power	P _{max}	5	mW	
Reverse voltage	V _R	1	V	
Forward current	I _F	1	mA	
Operating temperature*	T _{op}	-40 - +85	°C	Coaxial package
Operating temperature*	T _{op}	-40 - +60	°C	DIL, BTF (T _{st} = 25°C)
Storage temperature	T _{stg}	-40 - +85	°C	
Soldering temperature	T _{sold}	260	°C	Max. 5 seconds

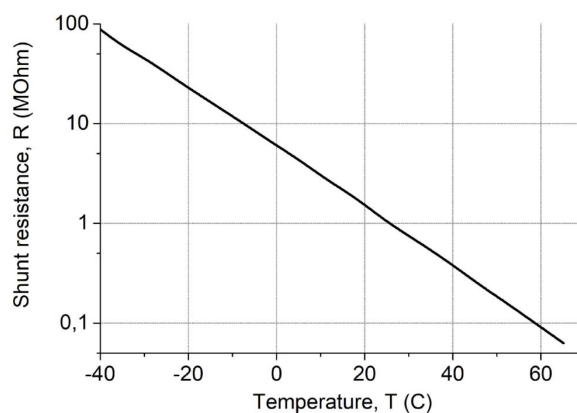
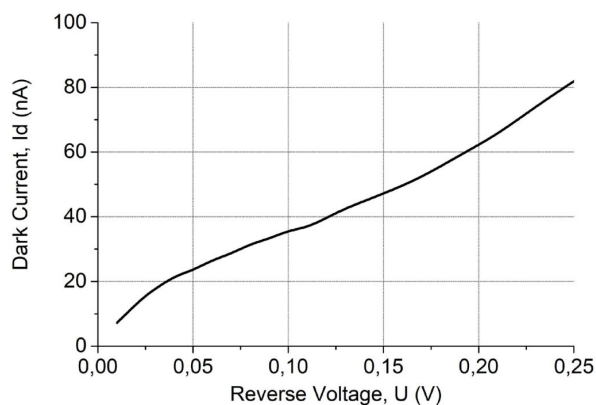
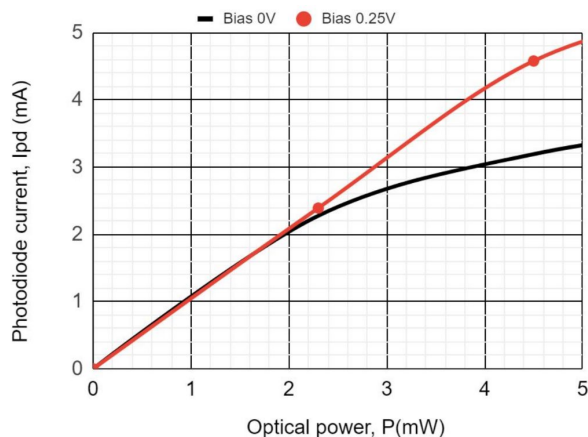
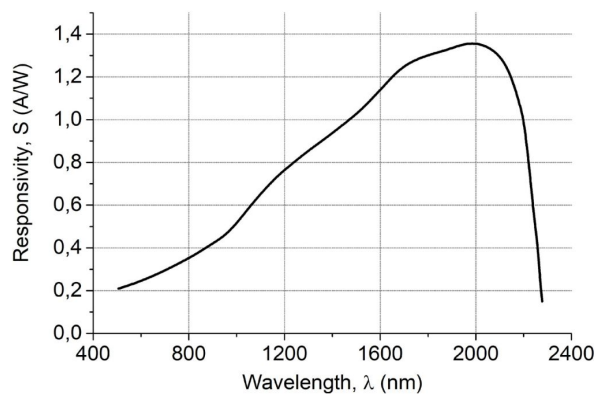
*Operating temperature is defined by the case temperature. It is necessary to ensure sufficient heat dissipation so that the module's maximum operating temperature is not exceeded. Operation at elevated temperatures reduces the lifetime of the laser diode.

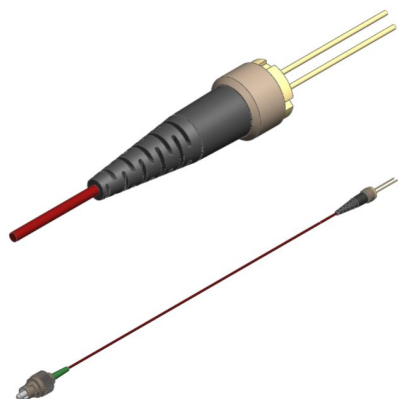
Operating temperature for the DIL, DILRAD and BTF case with TEC is defined for internal temperature stabilization at T_{st} = 25°C that corresponds to thermistor resistance R_t = 10 kOhm.

PD4B-250-P5-LC**ELECTRICAL-OPTICAL CHARACTERISTICS (T = 25 °C)**

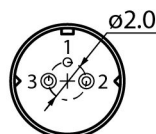
Parameter		MIN	TYP	MAX	Unit	Conditions
Operating wavelength	λ	500		2150	nm	
Wavelength cut-off	λ_c	2150			nm	
Wavelength of peak responsivity	λ_p	1850	1950	2050	nm	
R50/R30 Technology	Responsivity	R	0.33	0.41		$\lambda = 900 \text{ nm}, V_R = 0 \text{ V}$
			0.74	0.92		$\lambda = 1300 \text{ nm}, V_R = 0 \text{ V}$
			0.87	1.05		$\lambda = 1550 \text{ nm}, V_R = 0 \text{ V}$
				1.20		$\lambda = 1900 \text{ nm}, V_R = 0 \text{ V}$
	Return loss	RL	25	30		R50 (SM1, SM3, SM1950)
			13	16		R30 (MM5, MM6, MM105)
	Polarization dependent loss	PDL			0.15	dB
Operating voltage	V_R		0.25		V	
Dark current	I_d		0.2	1	μA	$V_R = 0.25 \text{ V}$
Shunt impedance		400	800		k Ω	$V_r = 10 \text{ mV}$
Total capacitance	C_t		40		pF	$V_R = 0 \text{ V}$
Peak specific detectivity	D^*		2.9e11		$\text{cm} \sqrt{\text{Hz}} / \text{W}$	
Peak NEP	NEP		7.8e-14		$\text{W} / \sqrt{\text{Hz}}$	

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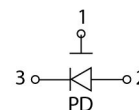
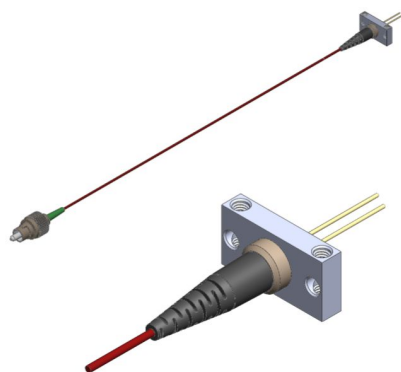
PD4B-250-P5-LC**COAX**

BACK VIEW

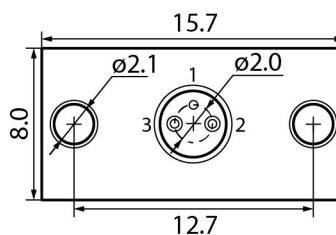


PINOUT

#7

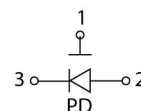
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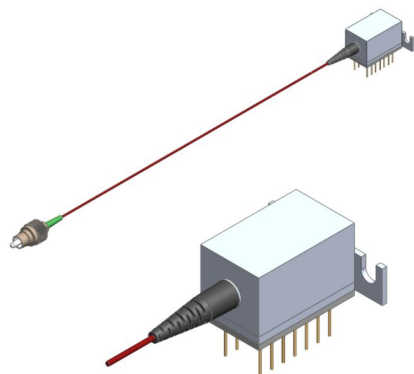
BACK VIEW



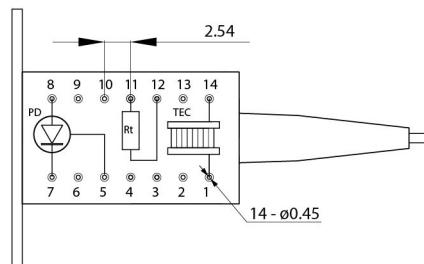
PINOUT

#7

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PD4B-250-P5-LC**DIL**

BOTTOM VIEW



PINOUT #5, #7

- 1.TEC Anode
- 2.-
- 3.-
- 4.-
- 5.PD Case
- 6.-
- 7.PD Cathode
- 8.PD Anode
- 9.-
- 10.-
- 11.Thermistor
- 12.Thermistor
- 13.-
- 14.TEC Cathode

TEC HP: $I_{\max} = 1.4A$, $U_{\max} = 3.9V$, $Q_{\max} = 3.3W$,
 $AC R = 2.0 \Omega$, $\Delta T_{\max} = 69 K$

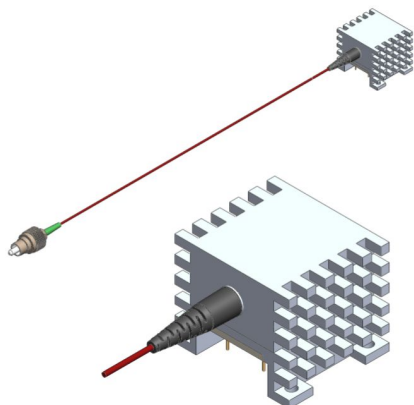
Thermistor:
 $R_t = 10 \cdot \exp(3600 \cdot (1/T[K] - 1/298)) \text{ k}\Omega$

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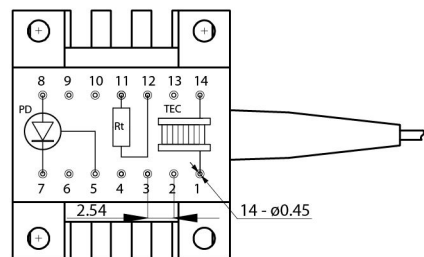
Drawing

3D model

Application Notes

**DILRAD**

BOTTOM VIEW



PINOUT #5, #7

- 1.TEC Anode
- 2.-
- 3.-
- 4.-
- 5.PD Case
- 6.-
- 7.PD Cathode
- 8.PD Anode
- 9.-
- 10.-
- 11.Thermistor
- 12.Thermistor
- 13.-
- 14.TEC Cathode

TEC HP: $I_{\max} = 1.4A$, $U_{\max} = 3.9V$, $Q_{\max} = 3.3W$,
 $AC R = 2.0 \Omega$, $\Delta T_{\max} = 69 K$

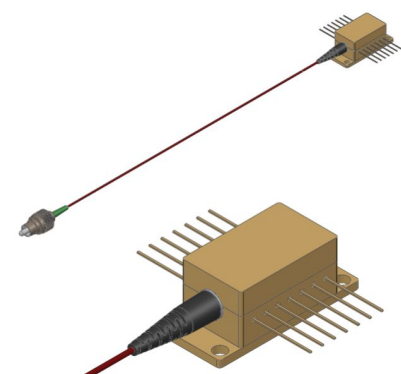
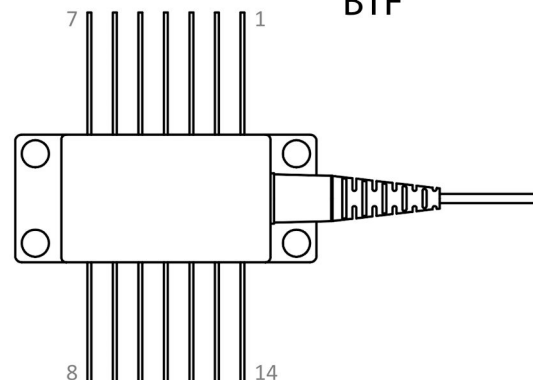
Thermistor:
 $R_t = 10 \cdot \exp(3600 \cdot (1/T[K] - 1/298)) \text{ k}\Omega$

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Drawing

3D model

Application Notes

**BTF**

1. TEC+
2. Thermistor
3. PD Anode
4. PD Cathode
5. Thermistor
6. NC
7. NC
8. NC
9. NC
10. NC
11. NC
12. NC
13. Case Ground
14. TEC -

TEC HP: $I_{\max} = 1.4A$, $U_{\max} = 3.9V$, $Q_{\max} = 3.3W$,
 $AC R = 2.0 \Omega$, $\Delta T_{\max} = 69 K$

Thermistor:
 $R_t = 10 \cdot \exp(3600 \cdot (1/T[K] - 1/298)) \text{ k}\Omega$

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3D model

Application Notes

PD4B-250-P5-LC

Characteristics, data, materials and structures specified in this datasheet are subject to change without notice. Please refer to the latest specification before use of the products.

Safety and handling cautions

1. Avoid smashing and burning of the module. Avoid storing and using the module in conditions where water, organic solvents or aggressive acids or bases may contact the module or where there is a possibility of exposure to corrosive gases, explosive gases, dust, salinity or other harsh conditions. The module should be disposed as special industrial waste.
2. Exceeding absolute maximum ratings even for a short time can cause permanent damage of the module.
3. The module is sensitive to and can be broken by ESD (static electricity).

Conflict Minerals Policy Statement

LD4B achieves business objectives and customer needs with social responsibility. We do not support or contribute to the violence and human rights violations associated with the mining of conflict minerals coming from Conflict Regions according to US "Dodd-Frank Act". When possible, our suppliers' conflict mineral statements are reviewed. We do not directly purchase Conflict Minerals from any source and do not knowingly procure any parts and products containing Conflict Minerals from Conflict Regions.

RoHS Compliance Statement

Restriction of Hazardous Substances (RoHS) directive (Directive 2011/65/EC amended with Directive (EU) 2015/863) is the directive aimed at reducing the harmful environmental impact of waste electrical equipment by restricting the use of known dangerous substances. Based on information received from our supply sources, LD4B hereby states that the banned substances listed in the RoHS directive are not found in the parts and materials used above the threshold level listed other than exceptions approved by the European Commission.

REACH Compliance Statement

Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) is a European Union regulation 1907/2006/EC that addresses the production and use of chemical substances, and their potential impacts on human health and the environment. Based on information received from our supply sources, LD4B hereby states compliance of the parts and materials used in manufacturing to REACH regulation. LD4B does not manufacture or import any substances or preparations as defined under REACH.